

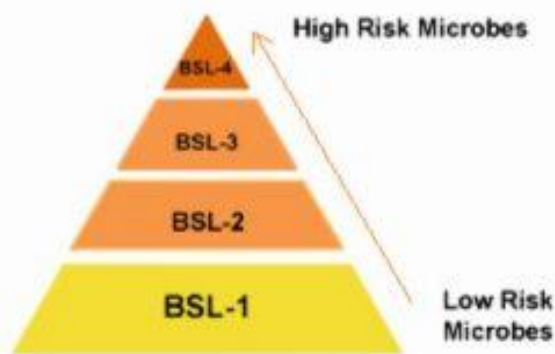
Biosafety and Biohazardous Waste

Biosafety

The CDC defines **biosafety** as “the application of safety precautions that reduce a laboratorian’s risk of exposure to a potentially infectious microbe and limit contamination of the work environment and, ultimately, the community.” The laboratory that you will be using for your experiments and procedures is a Biosafety level 2 (BSL-2) laboratory.

What does being a BSL-2 laboratory means to you and your safety?

The Centers for Disease Control and Prevention (CDC) and the National Institute of Health have established four levels of containment called Biosafety levels (BSLs). Each level has specific control measures for the containment of types of microbes and biological agents. The levels are assigned based on the evaluation of specific risks that the microbe or biological material may have on laboratory personnel and the community. The level of containment correlates with infectivity, the severity of disease, transmissibility, and the nature of the work conducted. Other risk factors that may also play in the determination of the BSL level include the origin of the microbe or agent and route of exposure.



Biosafety Levels (BSLs)

The CDC defines containment requirements for BSL-2 as the following:

“If you work in a lab that is designated a BSL-2, the microbes there pose moderate hazards to laboratorians and the environment. The microbes are typically indigenous and associated with diseases of varying severity. An example of a microbe that is typically worked with at a BSL-2 laboratory is *Staphylococcus aureus*.” The containment requirements that we are required to abide by including the following.

Laboratory practices

- Access to the laboratory is restricted when work is being conducted.

In our lab, this means that we do not allow individuals that are not enrolled or part of the teaching staff into the laboratory when we are conducting procedures.

Safety equipment

- Appropriate personal protective equipment (PPE) is worn, including lab coats and gloves.
- Eye protection and face shields:
 - Eye protection must be worn when staff or students are handling risk group 2 (RG2) organisms at the benchtops
 - Face shields must be used when performing procedures that may create a splash hazard
 - Eye protection and face shields are disinfected with an appropriate agent (e.g., BacDown®) at the end of each laboratory session.
- All procedures that can cause infection from aerosols or splashes are performed within a biological safety cabinet (BSC).
- An autoclave or an alternative method of decontamination is available for proper disposal.

In our lab, unless specified otherwise, you are required to wear clean scrubs (top and bottoms) or a long laboratory coat with long pants or a skirt along with closed-toed shoes with disposable gloves. You are responsible for the proper disposal of biological hazards into the appropriate waste containers (discussed later). You are required to disinfect your tables and equipment before leaving the laboratory. The laboratory staff will dispose of the biological waste for decontamination.

Facility construction

- The laboratory has self-closing doors.
 - A sink and eyewash are readily available.
- Every laboratory, including ours, regardless of the biosafety level follows the standard of microbiological practices. These practices include but are not limited to:
- Not eating, drinking, or applying cosmetics in the lab
 - Washing hands after working with infectious materials and before leaving the lab
 - Routinely decontaminating work surfaces

Biohazardous waste

In this laboratory, we will be working with BSL-2 agents as part of your learning experience. To ensure everyone's safety and enjoyable learning environment, we have a rigorous biohazard waste policy.

What is biohazardous waste?

Biohazardous waste, also called *infectious waste* or *biomedical waste*, is *any waste containing infectious materials or potentially infectious substances* such as blood. Of special concern are sharp wastes such as needles, blades, glass pipettes, and other wastes that can cause injury during handling. The waste reciprocals are labeled with the following red biohazard waste label.



Biohazard Waste Label

1. Sharps container

Each of your benches is equipped with a **sharps container**. Items that are sharp enough to puncture the skin and contaminated with unsterilized biological material should be placed in these bins.

Examples of items used in our lab that should be placed into the sharps container include:



Sharps Containers

- Glass Pasteur pipettes
- Glass slides
- Biologically contaminated broken glass

If they are overflowing or reaching capacity (3/4 full) please notify an instructor or teaching assistant.

2. Solid non-sharp biohazardous waste

Large biohazardous trash cans are located throughout the lab and are denoted by the large **orange** biohazardous waste sticker on their sides or lids. Lab consumables that have come in contact

with viable biological material that contains any laboratory material that is regarded as potentially infectious should be placed in the bins.

Examples of these types of items that should be placed in the biohazardous bins include:



Biohazardous Bins

- Gloves
- Culture flasks
- Plastic pipettes
- Pipette tips and well plates
- Wastes items contaminated with blood or other infectious or biological material

3. Liquid biohazardous waste

In some labs, we will be working with biohazardous liquids, mainly for **fecal** and **urine** procedures. On those days, the liquids should be placed in the liquid biohazard waste containers.

1. What are biological hazards or biohazards?

Biohazards include biological agents and toxins infectious to humans, animals, wildlife, or plants such as parasites, viruses, bacteria, fungi, prions and biologically active materials such as toxins, allergens, and venoms. In addition to causing disease in living organisms, biohazards can cause significant damage to the environment.

2. Why is research on biohazards important?

This provides the foundation for developing medical products and strategies to diagnose, treat, and prevent a wide range of diseases, whether those diseases emerge naturally or are deliberately introduced into a population through an act of bioterrorism.

3. How do we ensure that people, animals, and the environment are protected from biohazards in the laboratory?

People, animals, and the environment are protected from biohazards in the laboratory through the use of specific practices, training for laboratory staff, safety equipment, and specially designed buildings. The discipline which ensures that people, animals, and the environment are protected from biohazards in the laboratory is called laboratory biosafety.

4. What is biosafety?

Biosafety is a framework that describes the use of specific practices, training, safety equipment, and specially designed buildings to protect the worker, community, and environment from an accidental exposure or unintentional release of infectious agents and toxins.

It is also simply defined as the use of practices and principles to reduce the health-related risks associated with handling infectious agents, toxins and other biological hazards in a laboratory setting. Examples of such practices include:

- o biosafety cabinets;
- o personal protective equipment including masks, gloves, safety glasses, and lab coats;

- o hand washing; and
- o procedures for safe use of chemicals.

5. What are biosafety levels?

“Biosafety Levels” (BSLs) are designations applied to projects or activities conducted in laboratories in ascending order of containment based on the severity of the health-related risk associated with the work being conducted. The designations BSL-1, BSL-2, BSL-3, and BSL-4 describe the minimum safe work practices, specially designed buildings, and safety equipment required to conduct work on infectious agents, toxins, and other biological hazards. BSL-4 is the highest biosafety level. The appropriate BSL to be assigned to a project is determined by institutional biosafety committees (IBCs) or professionals, and reflects the specific combinations of specially designed buildings, safety equipment and safe work practices that laboratory workers must use. The BSL assigned to a project is based on a biological risk assessment.

6. What is laboratory biocontainment?

“Laboratory biocontainment” refers to the use of safety equipment and specially designed rooms and buildings which act as primary barriers to prevent the release of infectious agents, toxins, and other biological hazards into the environment. Examples of biocontainment barriers include, but are not limited to:

- o biosafety cabinets; and
- o sealed containers such as containment centrifuges and special animal caging systems.

7. What do the terms high and maximum containment mean?

“High containment” is a term used to describe laboratories or facilities that operate at Biosafety Level 3 (BSL-3) conditions. “Maximum containment” is a term used to describe laboratories or facilities that operate at BSL-4 conditions. Some documents use the term “high containment” to refer to both BSL-3 and BSL-4 facilities. The research activities that take place in high and maximum containment facilities include studies of high-risk infectious agents and toxins that can cause significant harm to humans, animals, or plants and can be transmitted by aerosol.

All biocontainment designations are based on the degree of the health-related risk associated with the work being conducted.

8. What is biorisk?

“Biorisk” is a combination of the likelihood of an exposure to an infectious agent, toxin, or biological hazard that will cause harm, and the consequence, or severity, of that harm if exposure does occur. The higher the combination of likelihood and consequence of a harmful event resulting from exposure, the greater the risk will be. Biorisk includes risks from:

- o accidental infection, toxic response, or allergic response;
- o accidental release to the community or environment; and
- o unauthorized access, theft, or misuse by an individual who intends to cause harm.

9. What is a biological risk assessment?

A biological risk assessment is a process that evaluates multiple factors to determine the risk to laboratory workers, the community, or the environment of working with an infectious agent, toxin, or other biological hazard. The biological risk assessment is used to determine the appropriate biosafety level for each project conducted within a laboratory. Understanding the nature of the infectious agent, toxin, or biological hazard and the way in which the infectious agent, toxin, or biological hazard causes disease is the first step in conducting a biological risk assessment.

10. What is laboratory biorisk management?

“Laboratory biorisk management” is a management system framework that describes a set of administrative practices that an institution can use to identify, monitor, and control the laboratory biosafety, biocontainment, and biosecurity aspects of its operation to create a safe and secure laboratory environment. Laboratory biorisk management is a useful tool to identify and implement appropriate laboratory safety and security measures.

11. Why do we need a culture of safety and responsibility in the laboratory?

A culture of safety and responsibility in the laboratory helps ensure safe and responsible behaviors and practices. Individual and organizational attitudes about safety will influence all aspects of laboratory conduct, including whether individuals will comply with safety and security measures and procedures, report biosafety or biosecurity concerns, respond appropriately to incidents; and communicate risks to each other and to managers in a timely and accurate manner. Every organization should strive to develop a culture of safety and responsibility that promotes transparent communication without fear of reprisal, encourages questions, and welcomes evaluation of its institutional practices.

Laboratory workers have the responsibility to report their concerns to their managers and the right to express their concerns without fear of reprisal. Similarly, managers have the responsibility to address the concerns that are raised. Both laboratory workers and managers must work to maintain the highest standard of safety and responsibility.

12. What is laboratory biosecurity?

Laboratory biosecurity refers to the protection, control of, and accountability for high consequence biological agents and toxins, and critical relevant biological materials and information within laboratories to prevent unauthorized possession, loss, theft, misuse, diversion, and accidental or intentional release.

14. What is the relationship between laboratory biosecurity and biosafety?

The fifth edition of the Biosafety in Microbiological and Biomedical Laboratories (BMBL) describes biosafety and biosecurity as "... related, but not identical, concepts. Biosafety programs reduce or eliminate exposure of individuals, the environment, and experimental materials to potentially hazardous infectious agents and toxins. Biosecurity on the other hand is to prevent loss, theft, or misuse of microorganisms, biological materials, and research-related information. This is accomplished by limiting access to facilities, research materials and information. While the objectives are different, biosafety and biosecurity measures are usually complementary.

15. What is meant by physical security in biosecurity?

Physical security encompasses the application of operational and security equipment, personnel and procedures used to protect facilities, and information, documents or material for preventing or responding to theft, sabotage, diversion, or other terrorist or criminal acts.